PART 1: Identify Network Information

- **What is your IP address?**
- Sagot: 192.168.0.104
- **What is the name of your network interface?**
- Sagot: wlan0
- **What is your default gateway?**
- Sagot: 192.168.0.1

PART 2: Test Network Connectivity

- **Did the request succeed? (Yes/No)**
- Sagot: Yes. Nakatanggap ng 4 received packets at 0% packet loss.
- **What is the average time?**
- Sagot: 51.415 ms

PART 3: DNS and Network Path

- **What IP address is returned?**
- Sagot: 202.91.163.13
- **How many hops does it take to reach the target?**
- Sagot: 13 hops (Dito natapos ang huling successful hop bago umabot sa target IP).

PART 4: Check Open Ports

- **List at least 2 open ports / What service is running on those ports?**
- Sagot: Walang lumabas na open ports sa output ng ss -tulnp. Ibig sabihin, ligtas ang Kali Linux laptop at walang active server/services na naka-run sa background na maaaring pasukan.

PART 5: Network Scanning

- **What ports are open? / What services are detected?**
- Sagot: "All 1000 scanned ports on 192.168.0.101 are in ignored states". Walang na-detect na open ports dahil ang in-scan na target (Windows 11 desktop) ay naka-on ang Firewall, kaya bina-block nito ang port scanning.

PART 6: Packet Capture

- **What interface will you use?**

- Sagot: wlan0
- **What kind of traffic did you see?**
- Sagot: Nakakita ng mga UDP packets, ARP requests, at DNS PTR queries.
- **Did you observe any IP addresses?**
- Sagot: Yes, nakita sa capture ang IP ng Kali (192.168.0.104), IP ng Router (192.168.0.1), Target PC (192.168.0.101), at isa pang device sa network (192.168.0.103).

FOCUS QUESTIONS

- **Why do some commands require sudo?**
  - Sagot: Ang sudo (Superuser Do) ay kailangan para mabigyan ng administrative o "root" privileges ang user. Importante ito sa mga command na direktang nagmo-monitor ng raw network interfaces o nagbabago ng system settings para maiwasan ang unatorisadong pag-access.
- **What is the importance of nmap in cybersecurity?**
  - Sagot: Ang nmap ay kritikal na tool para sa Network Discovery at Vulnerability Scanning. Nakakatulong ito sa mga security professionals na malaman kung anong mga device ang konektado sa network at kung anong mga ports o services ang bukas na posibleng maging kahinaan laban sa mga attackers.
- **How can tcpdump help in monitoring attacks?**
  - Sagot: Ginagamit ang tcpdump upang i-capture at suriin ang real-time data packets na dumadaan sa network. Makikita rito kung mayroong kahina-hinalang trapiko, unauthorized connections, o data exfiltration na palatandaan ng cyber attack.

BONUS CHALLENGE

- **nmap -sV target (What additional information did you get?)**
- Sagot: Ang -sV command ay nagde-detect ng "Service Versions" para malaman ang eksaktong application na naka-run sa isang port. Sa kaso ng output, nag-perform ito ng "Service detection", subalit hindi ito nakakuha ng versions dahil naka-block ang ports ng firewall.